



EDUCATION

Year	Degree	Institute	Marks
2024 - 2028	B. Tech in Mechanical Engineering	Indian Institute of Technology Madras	9.03/10
2024	ISC (12 th Class)	Head Start Educational Academy	96%
2022	ICSE (10 th Class)	Head Start Educational Academy	89%

ACADEMIC ACHIEVEMENTS & SKILLS

Technical Skills / Relevant Courses	- CAD, FEA, CFD, ROS2, AI/ML Implementation, MATLAB, Java, Simulink - Thermodynamics, Fluid Mechanics, Basic Electrical Engineering, Kinematics and Dynamics of Machinery, Measurements Instrumentation & Control, Linear Algebra, Python Data Structures & Algorithms (ongoing).	2026
Competitive Exams	JEE Mains (Score: 273, Rank: 3007)	2024
	JEE Advance (Score: 205, Rank: 2130)	2024

TECHNICAL PROJECTS & RESEARCH INTERNSHIPS

HiRo^[7] Lab Intern IISC Prof. Ravi Prakash <i>*Ongoing</i>	- Modifying, training and running inference on video world models like Gaussian World Model, DINOv3, pi0.5, - Working on novel ideas in Closed Loop Systems for World Models to make them more <i>geometry aware</i>. - Innovated a memory-retrieval term for WorldPack, improving SSIM ~3.5% /LPIPS ~3% over baseline on LoopNav. - Prototyped RoPE, trajectory packing and geometric/semantic attention conditioning on the NWM backbone.	
National Winner Caterpillar Challenge 2026 (Mar – June)	- Headed a team of 3, developed a python based Multi - Agent dump truck simulator for autonomous dump packing, winning Caterpillar's national contest with upto 7500+ competitors through a robust, deployable solution. - Solved the task selection and optimal assignment problem using a Heuristic, Hungarian Algorithm, MCTS^[5]. - Integrated vehicle dynamics and path planning to optimize complex truck reverse maneuvers into dump targets.	
Hub Motor Cooling Jacket Design Formula Student 2025 (July – Nov)	- Designed a cooling jacket for an electric hub motor to efficiently cool the motor with optimized pressure drop and maximized Heat Transfer Coefficient, using specific configurations of lattice structure designs. - Developed a MATLAB iterative solver that determines the overall head loss across the cooling loop of the car integrating it with results obtained from CFD Thermal analysis of certain cooling components. - Achieved improved accuracy in cooling simulations through multi-phase CFD modelling.	
Inter IIT Tech Aerodynamics 2025 (Oct – Nov)	- Worked on a LAT Aerospace problem statement focused on high-lift optimization for STOL aircraft. Performed extensive CFD simulations and research-driven analysis to test, compare, and propose unique aerodynamic solutions. - Developed toolpath to manufacture a modified airfoil configuration integrated with USB^[4] for wind tunnel testing.	
Composite Structurals Formula Student 2025 (Oct – Nov)	- Used the ACP module (Ansys Composite PrepPost) for failure analysis of the car's front wing. - Ensured deflection for specific load cases was within FSAE regulations through structural optimization of components. Validated simulation results with real life deflection testing.	
MIC (ME2400) Course Project 2026 (Apr – May)	- Developed an automated medicine dispenser by designing the working pipeline, selecting and integrating sensors/actuators, developing CAD models, assembling the system, and programming its functionality. Awarded as the <i>best project</i> in our batch by the former HOD, Prof. Chandramouli.	

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

Aerodynamics and Cooling Engineer Formula Student 2025 (Apr – Dec)	- Part of the engineering team responsible for aerodynamic component design and manufacturing. Gained extensive hands-on knowledge of aerodynamics, cooling strategies, vehicle dynamics & power electronics integration. - Looked after precise procurement, manufacturing, and assembly of all Carbon Fiber components.	
Business Plan Presentation Runners Up RFR^[1] 2025 (Jun – Sept)	- Was part of the team which ideated and planned out an entire business plan for the Business Plan Presentation event at Formula Bharat 2026, eventually securing 2nd place out of 40+ teams across India. - Worked on the entire market analysis of the fictional company and worked on optimizing the Supply Chain. Gained experience in topics like TCO analysis, QA, competitive analysis, market sizing, CAC estimation. - Developed a pitch deck, market numbers and finance sheets for the fictional company.	

EXTRACURRICULAR ACTIVITIES

Inter – IIT Music Contingent Head <i>(Piano)</i>	- Led a team of 8 musicians as Music Director of Stagecoach: Drama Club's highest budget annual event. - Ranked 2nd in the Band Competition among all the semi-professional bands in the Instincts 26' fest. - Represented IIT-M at Inter-IIT Cultural Meet 8.0 as Lead Keyboardist, placing 4th among 23 IITs. - Successfully completed all graded exams in Piano ABRSM and achieved merit in Grade 8.	2025
Oratory Contingent <i>(Debate)</i>	- Participated in multiple inter college offline and online Parliamentary debates including National Law School Bangalore (NLSIU Bangalore) Parliamentary Debate 2025, IITMPD 2025, Gambit 2025. - Competed in debates on Criminal Justice reform, macroeconomics, International Relations, & Public Policy.	2025
Chess	Obtained a FIDE rating of 1175 at the age of 8, participated in under 9 nationals and ranked 35th.	2015